

Sprint 4 - Technical Demo and Report

Purpose

The purpose of this sprint is to present your findings and justify your design decisions to stakeholders. Students will demo the prototypes and write a report on their usability study.

Group Work Requirement

Please read through the group work requirements here: [link](#). Duplicate the whiteboard and document your project in it.

Deliverables (3 + 4)

Part 1 (Demo)

1. Turn in the Technical Demo Directions quiz in Canvas by the due date (*individual submission*).
2. Submit the slide deck for the Technical Demo in Canvas (*group submission*).
3. Technical Demo due in class (*group submission*).

Part 2 (Report)

1. Turn in the Technical Report Directions quiz in Canvas by the due date (*individual submission*).
2. Submit the draft of Technical Report for peer review in Canvas under Discussions (*individual submission*).
3. Turn in the final Technical Report in Canvas by the due date (*group submission*).
4. Turn in the accessible link to FigJam whiteboard used to document your sprints through the semester (*group submission*).

page. To prepare for the quiz, read all the requirements for the assignment carefully and take notes. You may use your handwritten notes during the quiz, no other resources are allowed. Complete the quiz in one sitting, as you will have only one attempt. If you face a technical difficulty, then please take a clear screenshot of the error and send it to the instructor and the head TA.

Head to Canvas to take the quiz.

Tech Report Assignment

Over the course of this semester, you completed several sprints from selecting an existing product to demoing your solutions to the class. Now, your final major deliverable for Part 2 of Sprint 4 is a comprehensive technical report detailing the entire project lifecycle.

Written in a formal tone, the report will be submitted as a website. This format will allow you to integrate text, data visualizations, embedded Figma prototypes, and other media/content. Use any free website application such as Google Sites, Github Pages, Wordpress, etc.

The **audience** of your report is the hypothetical product management and engineering team of your chosen product. The **purpose** is to document your methodology, clearly present usability data, and justify your decisions to persuade stakeholders to adopt your design.

The website should contain 3 webpages, Home, Data, and Recommendations. The homepage should contain the **title** of your study/project, **names** of all the authors, and a summary. Add a banner image at the top of the page to go with the title. **Summary** should be ~300 words, briefly stating the product you analyzed, the core problems discovered during testing, the solution you prototyped, and the key takeaway or recommendations.

Next, write and display the following section on the homepage.

Introduction (3-4 paragraphs) -- Provide the context for your project. Introduce the existing product. Define the target user demographic for this product. Why did your team select it? What were your initial hypotheses about its usability flaws?

Next, write and display the following sections on the Data page. This page proves that your design changes are based on data, not opinions.

Methodology (3-4 paragraphs) -- Briefly explain how you tested the existing product (ex. number of participants, demographics, tasks assigned, pre and post test questions, tools used, metrics tracked, etc.).

Findings and Analysis -- In the findings subsection (5 paragraphs), detail the top 4-5 usability issues discovered. You must include at least 4 data visualizations (ex. bar chart of task success rate, line graph of task that took longest, etc.) and 1 table that contains at least 4 direct quotes from your test participants that highlight their frustration or confusion. Do not create any visuals on user demographics as we are only concerned with the product

changes are based on the evidence gathered from the usability tests. Use quantitative and qualitative data and name specific UI components. Refer to the visuals from the findings subsection if applicable. For example: As seen in the line graph above, because 80% of users failed to find the 'checkout' button due to its placement..., our prototype relocates this to the top right header....".

Do not just say users did not like it, instead provide data. Another example: ...users struggled significantly while searching for courses. During task #3, finding a 300-level elective, 4 out of 5 participants failed to complete the task within 1 minute. Screen recordings revealed that users were unaware the search filters existed, as they were hidden behind a generic icon. Participant 3 stated, 'I didn't know I could filter by semester, so I just kept scrolling through hundreds of classes'.

Finally, write and display the following sections on the Recommendations page.

Recommendations (4-5 paragraphs) -- Provide a short list of 4-5 major design changes your study recommends. Compare your designs to the existing product. Discuss the benefits, trade-offs, etc. offered by your designs. Focus on how the redesign improves efficiency, reduces user error, or increases accessibility. Connect the design changes to usability attributes. Include at least 4 "Before and After" visual comparisons (screenshots) of the existing product and your prototype. At the end of this section embed the accessible link to the high fidelity Figma prototype. Ex. "At the following link you will find our fully interactive prototype which..."

Conclusion (2-3 paragraphs) -- Wrap-up the report by reflecting on the design process and the next steps. Provide a final summary of the project's value. Reflect on what you and your team learned about the intersection of UI design and user behavior. Include a hand-off to the engineering team of your chosen product. What would you say to them for seamless implementation of your design?

Format

Use good design, color scheme, and font for your website. Use images, design elements, informative headings (H1, H2, H3), bold text, etc. to make the website readable and attractive. Keep related text and images **closely aligned** to form a single cohesive visual unit. All images/visuals must have captions and alt-text for accessibility. Maintain a professional, objective, and analytical tone. **Don'ts:** Do not have large walls of text. Do not place content and images in a zigzag layout.

Submission

Ensure your website, Figma link, and any other links in the submission are published and set to "Public" or "Anyone with the link can view." Submit the link to your website to Canvas by the due date. Test your link in an incognito browser before submitting. Also submit a PDF version of your report/website.



words, b) a list labeled Strengths with at least 3 items, and c) a list labeled Improvements with at least three items. Strengths and improvements should be based on directions of the assignment, rubric, and lecture material.

FigJam Board Submission

Throughout the semester, your team used a FigJam board as a central hub for project management, brainstorming, and record-keeping. As we are in the last sprint, you will do a final "clean-up" of this board and submit it. This serves as your official project archive, proving that your team collaborated effectively and followed sprint methodologies over the entire project lifecycle.

Design and format (10)

Site contains required pages (Home, Data, Recs). High visual quality (font, color scheme). No walls of text or zigzag layouts. All visuals have captions and alt-text. Tone is formal and analytical. Accessible link provided.

Homepage (10)

Banner image included. Summary (~300 words) concisely covers product, problems, solution, and takeaways. Intro (3-4 paragraphs) clearly defines context, why the product was chosen, initial hypotheses, and target demographic.

Data -- Methods and Findings (10)

Methods (3-4 paragraphs) clearly explain test details. Findings (5 paragraphs) detail top 4-5 issues. Includes at least 4 data visuals (no user demographics) and 1 table with 4 direct user quotes.

Data -- Analysis (10)

Analysis (5 paragraphs) explicitly justifies design changes using testing evidence. Accurately references quantitative/qualitative data and specific UI components. Links text to visuals effectively (e.g., "As seen in the graph...").

Recommendations and Conclusion (10)

Recs (4-5 paragraphs) list 4-5 changes links to usability attributes, benefits, etc.. Includes at least 4 "Before & After" screenshots. Accessible Figma link embedded at end. Conclusion reflects on UI/user behavior and includes a clear hand-off to engineering.

Design and format

Minor formatting errors. Missing 1 page or combined pages incorrectly. Some missing alt-text/captions. Occasional informal tone or minor zigzag/wall-of-text layout issues.

Homepage

Missing banner image. Summary or Intro is slightly too brief or long. Missing 1-2 elements (e.g., forgot to state target demographic or initial hypotheses).

Data -- Methods and Findings

Methods are vague. Findings list issues but are missing 1-2 data visualizations or based on user demographics or the quote table contains fewer than 4 quotes. Visuals lack clarity.

Data -- Analysis

Justifications rely somewhat on opinions rather than the data provided. Vague references to UI components. Fails to explicitly link the analysis text to the findings.

Recommendations and Conclusion

Missing 1-2 "Before/After" screenshots. Trade-offs/benefits are vague. Figma link works but is hard to find. Conclusion missing either the reflection or the engineering hand-off.

Design and format

Poorly organized, hard to read, or link requires login, or PDF submission missing. Major layout issues (heavy text walls, zigzag layouts). Lacks professional tone. No alt-text used.

Homepage

Missing Summary or Introduction entirely. Lacks clear explanation of the product, problems, or demographics. Word counts heavily missed.

Data -- Methods and Findings

Methodology is missing/unclear. Major missing visual elements (no charts or no quote table). Does not detail 4-5 usability issues.

Data -- Analysis

Relies heavily on opinions (e.g., "users didn't like it"). Fails to use collected quantitative/qualitative data to justify prototype design choices.

Recommendations and Conclusion

Missing "Before/After" screenshots. No working Figma link provided. Conclusion is missing or fails to include engineering hand-off.

Team Info and Charter (3)

Board includes team member information and a completed, detailed Team Charter (working agreements, communication plans, etc).

Project Tracker (3)

Board includes a clearly utilized project tracker (e.g., Kanban board, checklist, or timeline) showing task assignments and status.

Agenda and Minutes (3)

Board contains documented agendas and meeting minutes for team meetings held throughout the project lifecycle.

Sprint Info (3)

Board clearly shows evidence of work/documentation organized by Sprints 1 through 4 (e.g., labeled sections, sprint goals, artifacts).

Version History and Accessibility (3)

Board is clearly used through the semester as evidenced by the version history. Easy to navigate and submitted with a working public link.

*Incomplete**Team Info and Charter (0-1)*

Team charter is blank, significantly incomplete, or missing team member information.

Project Tracker (0-1)

Project tracker is missing, completely blank, or shows no evidence of use during the semester.

Agenda and Minutes (0-1)

Agendas and minutes are missing, very sparse, or fail to show what was discussed/assigned.

Sprint Info (0-1)

Missing documentation for multiple sprints, or work is not labeled/organized by sprint phases.

Version History and Accessibility (0)

Board was completed last minute or mostly in the second part of Sprint 4 and/or link restricted/requires login.

Tech Demo Directions Quiz

As the name suggest, this quiz is based on the demo assignment directions found on this page. To prepare for the quiz, read all the requirements/rubric for the demo carefully and take notes. You may use your handwritten notes during the quiz, no other resources are allowed. Complete the quiz in one sitting, as you will have only one attempt. If you face a technical difficulty, then please take a clear screenshot of the error and send it to the instructor and the head TA.

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Tech Demo In Class

Introduction (~1 min)

- Introduce yourself (group number and names of all group members so that the TAs can make a note).
- Provide context and background on your product. Name the product and what is the business? What does the product do?
- Discuss the details of your usability study. How many participants tested the product, in-person or remote usability testing, tools you used, etc.

Body (~3 mins) This is the main part of the demo.

- Discuss the findings of the usability study. Participant feedback on the post-test questions, most problematic tasks/features, most liked tasks/features, etc.
- Present your Hi-Fi prototype and discuss how it tries to solve the problems faced by the participants.
- Justify your design decisions for each of the critical screens/tasks in your prototype.

Conclusion (~1 min)

- Signal end of demo and wrap-up by restating the main points (product name, usability study highlights, key design decisions, etc.).
- Discuss next steps if any and thank your audience.

Delivery

- Demo should be minimum 5 minutes, max 6. When you come to the podium, first start the timer, say your names and group number clearly.
- You may use note cards but are not allowed to use your phone while presenting.
-  Maintain eye contact with the audience and do not just read.
- All group members should have somewhat equal talking time. Please practice your demo

Students will be contacted in a random order in the classroom. If your group number is called and you are not in class, you will not receive credit for the assignment.

Visual Aid // Slide Deck

- Create your own background for the slides; do not use any templates or themes.
- Include the link to your Hi-Fi prototype in the slides.
- Slides should not be text-intensive. You want to include at the most 2-3 bullet points per slide.
- Include any other appropriate graphics such as screenshots, visuals, icons, or other design elements to make slides appealing.

Tech Demo Lecture

Sample Slide Design

Exceeds Expectations

Introduction (5)

Clearly introduces all members & group number. Provides strong context/background of the product. Clearly details the usability study (participants, setting, tools).

Body (18-20)

Thoroughly discusses usability findings (feedback, problematic/liked tasks). Clearly presents the Hi-Fi prototype and specifically shows how it solves user problems. Provides strong, logical justifications for critical design decisions.

Conclusion (5)

Effectively signals the end of the demo. Concisely restates main points (product name, study highlights, key decisions). Mentions next steps and thanks the audience.

Delivery (9-10)

Strictly 5-6 minutes; timer started. Excellent eye contact (no reading from slides). Equal talking time for all members. Note cards used appropriately (no phones).

Visual aid (9-10)

Custom background used (no templates). Hi-Fi link included. Highly scannable (max 2-3 bullets per slide). Excellent use of screenshots, icons, and graphics.

Introduction (3-4)

Misses minor details (e.g., forgets group number or a usability study detail). Context is provided but could be clearer.

Body (13-17)

Discusses findings but lacks depth. Presents the prototype but connection to user problems is vague. Design justifications are present but surface-level.

Conclusion (3-4)

Wrap-up is slightly abrupt. Misses one element, such as restating key design decisions or discussing next steps.

Delivery (6-8)

Slightly under/over time limit. Some reading from slides/cards. Talking time is somewhat uneven.

Visual aid (6-8)

Custom background is plain or slightly distracting. Text is a bit heavy in places. Graphics are used but could be more impactful.

Introduction (0-2)

Fails to introduce the team properly. Lacks clear product context or omits significant details about the usability study.

Body (0-12)

Skips major usability findings. Prototype presentation is rushed or disjointed. Fails to justify design decisions or connect them to user testing.

Conclusion (0-2)

Ends abruptly without a summary. Fails to outline next steps or acknowledge the audience.

Delivery (0-5)

Significantly off the time limit. Heavy reliance on reading; lack of eye contact. One or two members dominate the presentation. Used a phone.

Visual aid (0-5)

Used a pre-made template. Missing Hi-Fi link. Text-intensive slides. Lacks relevant graphics or visual appeal.